Alexander Ryabchenko

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Profile

PhD student at the University of Toronto and the Vector Institute, researching the statistical and computational foundations of modern machine learning, including LLM post-training, reinforcement learning, and optimization.

Education

Doctor of Philosophy (PhD), Statistical Sciences 2026 – Present
University of Toronto *Toronto, ON*
- Advisors: Murat A. Erdogdu, Wenlong Mou, Stanislav Volgushev.

Honours Bachelor of Science (HBSc), Mathematics and Computer Science 2021 – 2025
University of Toronto *Toronto, ON*
- Cumulative GPA: 4.00/4.00 (Average grade: 98.0%).
- Governor General's Silver Medal (highest academic standing), John Black Aird Scholarship (top student).

Publications

- **A.R.**, Jian Qian, Wenlong Mou. Adaptively Incorporating Directional Hints into Zeroth-Order Optimization. To appear in *Advances in Neural Information Processing Systems* (NeurIPS), 2026. arXiv:2609.08277.
- **A.R.**, Idan Attias, Daniel M. Roy. Capacity-Constrained Online Convex Optimization with Delayed Feedback. To appear in *Advances in Neural Information Processing Systems* (NeurIPS), 2026. arXiv:2606.11711.
- **A.R.**, Idan Attias, Daniel M. Roy. A Reduction from Delayed to Immediate Feedback for Online Convex Optimization with Improved Guarantees. To appear in NeurIPS 2026. arXiv:2602.02634.
- **A.R.**, Wenlong Mou. Reinforcement Learning with Action-Triggered Observations. In the *Proceedings of the 43rd International Conference on Machine Learning* (ICML 2026), pp. 106402–106430. arXiv:2510.02149.
- **A.R.**, Idan Attias, Daniel M. Roy. Capacity-Constrained Online Learning with Delays: Scheduling Frameworks and Regret Trade-offs. In the *Proceedings of the 38th Conference on Learning Theory* (COLT 2025), pp. 4973-5014. Oral presentation. arXiv:2503.19856.

Preprints

- Idan Attias[*], Orin Levy[*], **A.R.**[*], Yishay Mansour, Uri Stemmer. Vanilla Policy Optimization Is Both Optimal and Differentially Private for Stochastic Contextual Bandits. arXiv:2609.33888.
- Emmanuel Esposito[*], Andrew Jacobsen[*], Hao Qiu[*], **A.R.**[*], Mengxiao Zhang[*]. Optimal Learning in Games with Arbitrary Delays: Best-of-Both-Worlds for Self-Play and Adversarial Opponents. arXiv:2611.07773.

[*]Equal contribution; alphabetical order.

Awards

Connaught International Doctoral Scholarship (50,000 CAD) 2026 – 2031

Governor General's Silver Medal 2025
- Canada's most prestigious academic award, established in 1873 and awarded to the undergraduate student with the highest academic standing upon graduation.

John Black Aird Scholarship 2025
- Award recognizing the top graduating student at the University of Toronto.

E. A. Robinson Medal in Sciences 2025
- Medal recognizing the top graduating science student at the University of Toronto.

Vector Institute Research Grants (30,000 CAD total; two-time recipient) 2023 – 2024

University of Toronto International Scholar Award (180,000 CAD) 2021 – 2025

Undergraduate Scholarships & Research Awards at the University of Toronto 2021 – 2025
- MCS Research Award (7,500 CAD)
- Joseph Alfred Whealy In-Course Scholarship (6,000 CAD total; four-time recipient)
- Peter Fantham Memorial Scholarship in Mathematics (2,000 CAD)
- Doris Geddes Scholarship in Mathematics (1,000 CAD)

Research Experience

Graduate Student Researcher 2026 – Present
Vector Institute *Toronto, ON*

- Working on the theoretical foundations of modern AI: LLM post-training, scaling laws, optimization.
- Previously completed undergraduate Vector Institute Research Internships (2023, 2024).

Research Assistant 2022 – 2026
University of Toronto, Department of Statistical Sciences *Toronto, ON*

- Studied the mathematical foundations of modern machine learning, with projects spanning feature learning in neural networks and high-dimensional statistical estimation.
- Completed the MCS Summer Research Internship (2022).

Visiting Researcher Spring 2023
Sunnybrook Health Sciences Centre *Toronto, ON*

- Improved Swin UNETR for 3D segmentation in cancer and musculoskeletal research using dilated attention and a custom loss, outperforming the standard architecture in ablation studies.
- Released the resulting implementation as an open-source codebase and presented the results at the Sunnybrook Orthopaedic Biomechanics Laboratory seminar.

Technical Skills

Programming: Python, C/C++, Bash, Julia.
Machine Learning: PyTorch, JAX, Hugging Face Transformers.
Development Tools: Git, Linux, Slurm, Docker.

Workshops & Talks

A Reduction from Delayed to Immediate Feedback for Online Convex Optimization. Poster.

- *Learning Theory Summer School & Workshop* (LTSS 2026), Copenhagen, Denmark (Jun 2026).

Reinforcement Learning with Action-Triggered Observations. Seminar Talk.

- *UofT Statistical Sciences Research Program* (UTSSRP 2026), Toronto, Canada (Jun 2026).

Capacity-Constrained Online Learning with Delays. Poster.

- *Vector AI/ML "Remarkable 2026" Conference*, Toronto, Canada (Feb 2026).
- *Southern Ontario Learning Theory Workshop* (SOLT 2025), Waterloo, Canada (Nov 2025).

Academic Service & Teaching

Conference Reviewer 2026 – Present

- International Conference on Learning Representations (ICLR): 2027.
- Advances in Neural Information Processing Systems (NeurIPS): 2026 (**Top Reviewer**).
- Conference on Learning Theory (COLT): 2026.

Teaching Assistant Winter 2025
University of Toronto *Toronto, ON*

- CSC311, Introduction to Machine Learning.
- CSC363, Computational Complexity and Computability.